

Hypergraph partitioning

Netlists are hypergraphs: one net can touch many cells

The idea

- Optimize hyperedge cut, not only pairwise edge cut
- Multi-pin nets dominate affinity in real designs
- Use clique expansion only for drawing or as a heuristic substrate

Pseudocode

- Hypergraph cut counts a net once if it spans more than one part
- Pseudocode is a span check per hyperedge
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- On the starter hypergraph, golden ABC versus DE leaves a single cut net
- Expanding to a clique can tell a different numeric story, report which model you used

Algorithm sketch — try these

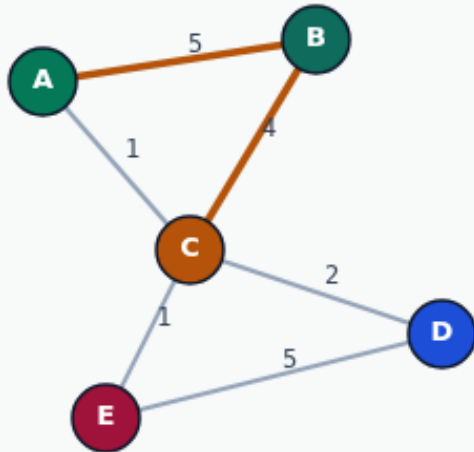
```
INPUT: hyperedges  $e=\{pins...\}$ ,  $side[]$   
OUTPUT:  $hyperedge\_cut$   
 $cut \leftarrow$  count edges with  $|\{side[p]: p \in e\}| > 1$   
(optional) expand to pairwise clique for KL/FM  
GOLDEN ABC|DE: one cut net on starter H  
pairwise clique expansion can differ
```

Nets are hyperedges

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Nets are hyperedges

Step 1 / 5 · nets · module03-05-hypergraph-partition



Explanation

TINY_HYPERGRAPH has four nets: $n1=\{A,B,C\}$ $w=3$, $n2=\{D,E\}$, $n3=\{C,D\}$, $n4=\{A,B\}$. Cut counts whole nets spanning ≥ 2 sides — not pairwise edges alone.

- Hyperedge cut if pins span sides
- $n1$ pulls ABC together
- Drawing uses clique expansion

4 hyperedges
5 nodes

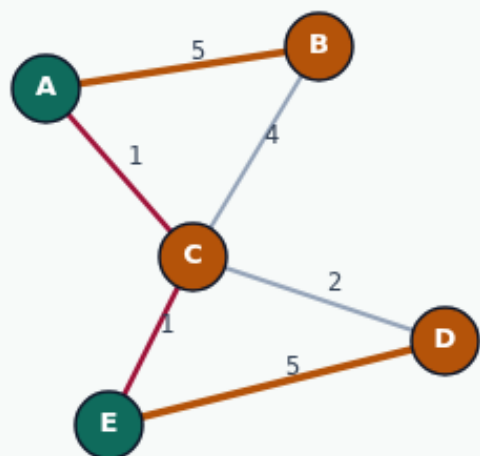
Nets are hyperedges

Bad seed: hyperedge cut 6

HYPERGRAPH PARTITION

Bad seed: hyperedge cut 6

Step 2 / 5 · bad-seed · module03-05-hypergraph-partition



Explanation

BAD_SEED AE|BCD cuts nets n1(3), n2(2), and n4(1) — hyperedge cut 6. Same seed as graph labs, worse hyper objective.

- n1 spans A vs B,C
- n2 spans D,E vs B,C,D
- Pairwise cut $\neq$ hyper cut

seed: AE|BCD
hyperedge cut: 6

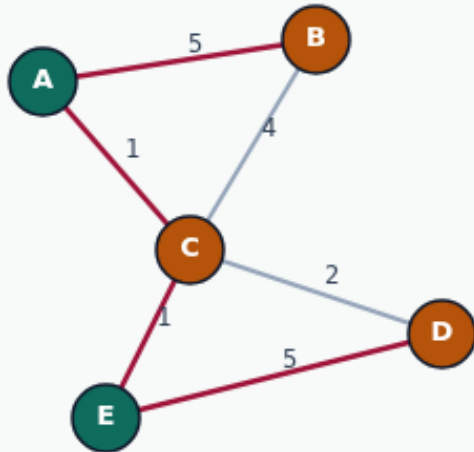
Bad seed: hyperedge cut 6

FM on clique expansion

HYPERGRAPH PARTITION

FM on clique expansion

Step 3 / 5 · fm-run · module03-05-hypergraph-partition



Explanation

Hypergraph FM clique-expands nets to pair edges, then runs standard FM from the bad seed. Objective reports hyperedge cut on the original nets.

- Clique expansion for moves
- Score hyperedge cut after moves
- Same FM kernel as graph labs

```
engine: FM on expansion  
seed hyper cut: 6
```

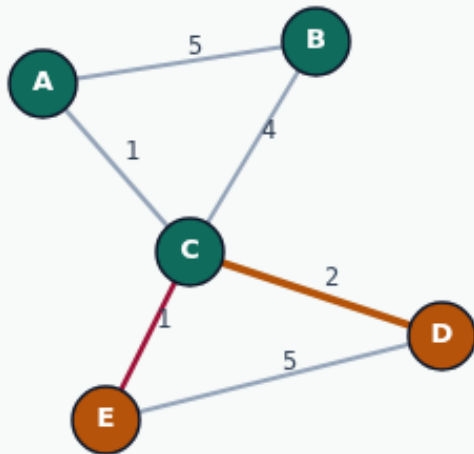
FM on clique expansion

Refined ABC|DE: hyper cut 1

HYPERGRAPH PARTITION

Refined ABC|DE: hyper cut 1

Step 4 / 5 · refined · module03-05-hypergraph-partition



Explanation

FM reaches ABC|DE. Only bridge net $n3=\{C,D\}$ crosses sides — hyperedge cut 1. $n1$, $n2$, $n4$ stay uncut.

- $n1$ ABC internal
- $n2$ DE internal
- $n3$ is the sole cut net

parts: ABC|DE
hyperedge cut: 1
pair cut: 1

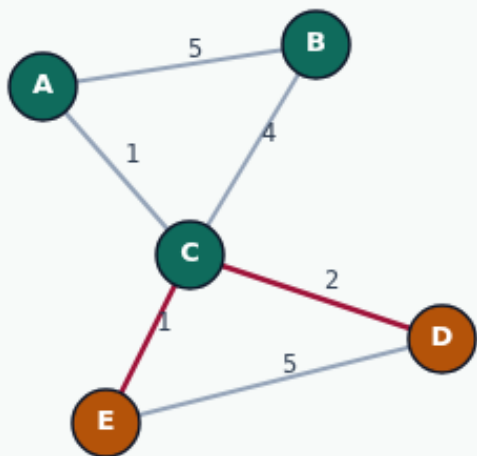
Refined ABC|DE: hyper cut 1

Why hypergraphs in partition

HYPERGRAPH PARTITION

Why hypergraphs in partition

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Explanation

Real netlists are hypergraphs. Modeling nets honestly changes the cut objective — clique expansion is for moves, hyper cut is for scoring.

- 6 → 1 hyper cut improvement
- Multi-pin nets dominate affinity
- Production tools use native hyper FM

Starter golden: hyper cut=1

Browser lab track

- In the browser lab track, open the **hypergraph-partition** lab from the tools shelf
- Load the starter graph, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse the tiny graph, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps
- For multilevel flows, verify coarsening before you blame the refiner

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module